Lesson 13 Teacher's Key: A Hit That Reacts

Scratch - Instructor Reference Only

VIDEO GAME CREATION

What This Lesson Teaches

Sequencing a visual reaction BEFORE finalizing game-state logic using Scratch's "wait" block, and spawning a fully independent visual effect via broadcast - the first lesson where the enemy's removal is deliberately delayed relative to when it was hit.

Level 1 - Full Worked Solution

- The Explosion sprite is a completely separate, self-contained script - it never touches score, lives, or wave variables, and its own "wait 0.2 seconds -> hide" lifecycle needs no help from the Enemy.
- "broadcast [explosion]" fires and moves on immediately - Scratch broadcasts never block the sender, which is exactly why the explosion can be triggered without disturbing the flash-then-wait sequence that follows it.
- "broadcast [enemy hit]" - the one line GameManager actually listens to for score/enemies_remaining/wave-clear - does not move at all. It stays exactly where Lesson 12 left it. The only structural change is adding "wait 0.1 seconds" AFTER it and moving "delete this clone" to after that wait.



Everything in the top box happens the instant the hit is detected. Only the enemy's own removal - the very last step - waits for the pause to finish.

Why This Structure

- Keeping the explosion as a broadcast-triggered, fully independent script (rather than something the Enemy waits on) means a slow or buggy explosion animation could never accidentally freeze the actual game logic - a good habit even at this small scale.
- This is the first lesson where 'wait' pauses something that MATTERS (delaying a state change) rather than just pacing a visual (Lesson 2's beep-beep) - but notice it's delaying the LEAST important thing happening in this branch (the enemy's own visual removal), not the GameManager broadcast that actually matters. That ordering is the whole lesson.

Level 2 / Level 3 Talking Points

- Ask students to trace, step by step, what happens between a bullet touching an enemy and the enemy actually disappearing now - sound, explosion, flash, broadcast [enemy hit] (GameManager updates score/wave immediately), THEN wait, THEN delete this clone. A good comprehension check for sequencing.
- For strong students: ask what would happen if "broadcast [enemy hit]" were accidentally moved to AFTER "wait 0.1 seconds" instead of staying before it - score and wave-clear would lag 0.1 seconds behind the visuals, which is exactly the mistake this lesson is designed to avoid.

Common Student Mistakes

- Moving "broadcast [enemy hit]" to AFTER the wait instead of leaving it where it was, which delays score/wave/win-check for no reason and contradicts this lesson's central rule.
- Forgetting to reset "set [color] effect to 0" somewhere in the enemy's spawn script, causing a freshly spawned enemy to start already tinted.
- Placing "delete this clone" BEFORE the wait instead of after, which defeats the whole point (the enemy would still vanish instantly, no visible flash).